

Hanbin Zheng

📍 Haining, Zhejiang ✉ hanbinz2@illinois.edu 🌐 hanbinzheng

Education

Dual degree program in Zhejiang University - University of Illinois Urbana-Champaign Institute:
University of Illinois Urbana-Champaign, Illinois, US Sep 2024 – Jun 2028
BS in Computer Engineering, GPA: 3.985/4.0
Zhejiang University, Zhejiang, China Sep 2024 – Jun 2028
BS in Electronic and Computer Engineering, GPA: 3.899/4.0

Awards and Honors

First Prize in ZJU Mathematical Modeling Competition	2025.06
Zhejiang Provincial Scholarship	2025.10
Zhejiang University Second Scholarship	2025.10
Third Prize in National RoboMaster University League (Shanghai, 3v3)	2026.04

Research Experience

Undergraduate Researcher, ZJUI Jun 2025 – Dec 2025
Advisor: **Prof. Xiangming Meng**

- **Flow Models Exploration** 📄
Conducted an in-depth study of Flow Matching generative modeling, implementing conditional image generation on MNIST and CIFAR-10 from scratch.
- **MeanFlow for Image Inverse Problem** 📄
Investigated one-step generative modeling with the recently proposed MeanFlow models, altered its initial distribution from Gaussian distributions to noisy images, achieving efficient one-step denoising.
- **GMFlow for Inverse Problem** 📄
Investigated Variational Flow Matching and Gaussian Mixture Flow Matching for image inverse problem; implemented 2D toy examples to validate our new algorithm.

Selected Projects

RoboMaster Hero Robot Electrical Control System 📄 Mar 2026 – Apr 2026

- **Embedded Control:** Developed real-time control system on STM32H723 for a steering wheel hero robot, managing high-frequency control loops and peripheral hardware integration.
- **Orientation Estimation:** Independently derived **quaternion kinematics** and implemented the **Mahony Filter** from scratch for IMU sensor fusion, achieving stable orientation tracking.

Mini GPT: Transformer From Scratch 📄 Nov 2025

- **Neural Architecture:** Built a decoder-only Transformer from scratch using Pytorch.
- **Mathematical Insight:** Analyzed dot-product attention as a kernel-based fusion mechanism, derived the interaction between sinusoidal positional embeddings and attention mechanisms, clarifying how relative distance is encoded into dot-product results.

Skills

Programming Languages: Python (PyTorch), C/C++, LaTeX
Robotics: STM32, Basic ROS2/ROS2 Control, Quaternion & Euler Angles, Soldering
Generative Modeling: Flow Matching, MeanFlow, Diffusion, GMFlow, UNet/Transformer Architecture
Development Tools: Git, Linux, Makefile, Vim/NeoVim, Emacs